Reviewer guide

Minimum-Cost Temporal Walk Profiles on Depth-Three In-Arborescences

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Companion prepared: 12 November 2026
Manuscript examined: working version of 5 November 2026, 27 pages

Read the manuscript | Detailed novelty comparison | Download the combined reviewer packet

The question for a reviewer

Does the manuscript prove a new, useful algorithmic guarantee for its precise restricted model, after accounting for temporal reachability, temporal path counting, and generalized deletion propagation?

The candidate contribution is the stated `2^{O(sqrt(R))} poly(L)` **algorithm and polynomial-space bound**, using a sufficient temporal state and partition-bounded configurations. Exact constraints, tree dynamic programming, and min-plus convolution are established ideas. The companion derives a GDP encoding and weighted ILP extension; the formulation and full-table output alone should not carry the novelty claim.

This author-side guide records a focused comparison and evidence map. It does not certify the proofs or originality. The manuscript remains a working paper; its frozen snapshot has not yet incorporated the new GDP analysis.

Plain-language contribution

Imagine timestamped transfers in a small tree leading to one account. Removing a transfer can break three-transfer chains while also destroying shorter chains that must survive. The algorithm returns the cheapest deletion set for each achievable pair of remaining counts. An upper limit and an exact target can have different answers.

The claim concerns computing this menu when little two-chain loss is allowed. It does not establish an operational transaction intervention, a changed fraud prediction, or reduced fraud.

Exact scope and notation

Item	Contract in the manuscript and reference solver
Input	An explicit finite event list. The original static graph is a directed in-arborescence toward one root, of depth at most three. Validate it before mandatory deletions.
Counts	`W2` and `W3` count strictly time-increasing event tuples ending at the root. Distinct event IDs remain distinct, including parallel events.
Choices	Optional event deletions, nonnegative integer costs, and mandatory keep/delete statuses. Mandatory deletion cost is charged once.
Baseline	After mandatory deletions, let `N2` and `N3` be the two counts and `T0 = max(T, 0)`. A quota above `N2` is infeasible.
Parameter	For a feasible quota, `R = N2 - T0`. The table uses exact loss `ell = N2 - W2` and exact surviving `W3`.
Output	Minimum total deletion cost and one attaining set of original event IDs for every attainable exact pair satisfying the quota.
Limit	A separate deletion-count cap in a weighted instance is not an additional coordinate of this table. Arbitrary other deletion-sensitive classifier features are outside its guarantee.

Source: manuscript Section 2 and Theorem 2.1, pp.2-3; reference solver.

Claim-to-evidence map

Page numbers below refer to the named 27-page manuscript, not a different draft. A code correspondence helps inspection; it does not prove a theorem.

Claim to examine	Manuscript location	Implementation or retained evidence
Complete minimum-cost exact profiles; subexponential dependence on `sqrt(R)`; polynomial space	Theorem 2.1, p.3; proof in Sections 3-5, pp.4-8	`solve_profiles`; complexity argument must be checked in the proof, not inferred from finite runs
Equal-degree final events are interchangeable under every continuation; cheapest canonical choices suffice; deletion charges are disjoint	Lemma 3.1 and Corollary 3.2, p.4	`_c_groups`, `_c_configurations` in the same solver
Branch configuration count is bounded by integer partitions; branches do not multiply this factor	Lemma 3.3, p.5; Section 5, pp.7-8	`_c_configurations`, `_branch_profile`, `_convolve`
Current scalar counts can lose information needed for later decisions	Proposition 3.4, pp.5-6	Ten-event illustration, expected output, and illustration checker
Chronological state and tie handling are sufficient; completed tables can be combined	Lemma 4.1 and Section 4, pp.6-7	`_child_profile`, `_convolve`; inspect equal-time event order and shared event identities
Unit-cost quota decision is W[1]-hard in deletion budget b, and NP-complete	Theorem 6.1, p.8; Appendix A, pp.16-22; Corollary A.7, p.22	Reduction in the manuscript. This is not a lower bound in preservation loss `R`
Count-based predicates can be optimized by a scan; an added dynamic amount feature has a hardness obstruction	Corollary 7.1, p.9; Proposition 7.2, pp.9-10	`select_profile` implements a predicate/total-cost selection layer; it does not replay a full fraud model
Fixed-deletion counting transfers to established forest counting; first-arrival summaries alone lose multiplicity	Proposition B.1 and Appendix B.2, p.23	Compare Counting Temporal Paths and the final reachability article
Finite correctness checks and synthetic transaction findings	Section 10, pp.13-16; Appendix D, pp.24-25	Mac correctness summary, batch export, recomputed totals

Useful reading route

Start with Theorem 2.1 and the normalization in Section 2. Then inspect Lemmas 3.1-3.3 and 4.1, followed by the complexity accounting in Section 5. Read the GDP construction in the comparison before assessing the novelty wording. Appendix A needs a separate reduction check; the experiments cannot validate that hardness proof.

For the worked illustration, distinguish an **exact loss** from a **lower quota**. The saved `global_loss_5` entry fixes `W2 = 5`. A query with `min_w2 = 5` permits larger surviving `W2` and can have a cheaper solution. These are different optimization requests.

What the experiments support

Separate correctness suite

The retained native Mac summary reports 375 histories, 1,863 threshold calls, 2,529 profile entries, and 54,076 selection queries. The exhaustive validator independently enumerates event tuples and legal deletion subsets rather than calling the solver's optimization helpers. This computational separation supports finite correctness checking; it is not a separate human proof review. See reproduction and evidence provenance.

Completed IBM synthetic-data batch

Outcome	Initial `W3 = 0`	Initial `W3 > 0`	Total
Empty context	179	0	179
Unsupported structure	125	155	280
Supported, above the 32-event solver cap	34	9	43
Fully profiled	454	68	522
All fixed queries	**792**	**232**	**1,024**

The positive-case conclusions concern **68 of 232** positive queries. Capped contexts were not proved infeasible. The original context was checked for structural eligibility; conflicting edges were not deleted to create an eligible tree.

For `U = floor(3 * B3 / 4)`, where `B3` is the initial three-event count, and unit optional deletion costs:

Two-event preservation	Upper bound `W3 <= U`, cost at most 3	Exact `W3 = U`, cost at most 3	Exact target unattainable at any cost
100%	68	13	55
95%	68	14	54
90%	68	15	53

These are repeated policies on the same 68 positive profiled contexts. The upper-bound target is always feasible without a cost limit by deleting all first-stage events; the observed result concerns the three-deletion limit. Only two queries changed exact attainability or minimum cost as preservation was relaxed.

Verification and interpretation limits

The native batch reports 935 witness checks. Of 563 completed numeric threshold tables, 528 agreed with exhaustive enumeration: 721 entries and 1,096,402 subset visits. The remaining 35 tables exceeded the 16-optional-event oracle limit; their 214 entries had witness checks without exhaustive optimality checks. No mismatch was reported.

The **89 positive-context tables** have `R` distribution `{0: 68, 1: 18, 2: 3}`. Across **all 563 tables**, the distribution is `{0: 522, 1: 35, 2: 3, 3: 2, 6: 1}`; the larger values occur only with initial `W3 = 0`. These data do not test scaling at large `R`.

The aggregate export lacks complete original contexts and native deletion witnesses. Reaggregation checks record consistency and totals, not a fresh replay. The cohort had already been studied, may contain overlapping contexts, and is not a held-out evaluation. Its 18,432 policy rows are not independent experiments. No fraud-accuracy or operational-effectiveness improvement is measured.

Sources: results guide, aggregate export, recomputed JSON, and frozen protocol. The later interactive explorer has separate fixture checks; they do not enlarge the manuscript's experimental sample.

Questions that would make a review actionable

1. **Correctness.** Is the final-event equivalence valid under every continuation, with equal timestamps, parallel IDs, forced choices, and nonnegative binary costs? If not, give the smallest counterexample and the failed lemma.
2. **Running time and space.** Does the charging argument bound every enumerated branch configuration by preservation loss, and does the processing order avoid multiplying partition factors across branches? Are witness recovery, integer arithmetic, and all retained tables polynomial in explicit input length?
3. **Prior-art implication.** Does a specific existing theorem imply Theorem 2.1 after a transformation? Please identify the theorem, give the encoding, and bound both its size and controlling parameter. GDP expressibility alone does not supply that complexity analysis.
4. **Alternative state or algorithm.** Can known query, ordered-incidence, or temporal-graph methods recover the result with the same or better guarantee? The direct first-arrival counterexample does not exclude richer constructions.
5. **Hardness.** Does Appendix A preserve graph depth, optional unit-cost events, both quotas, and the deletion budget in both directions? The manuscript uses `b` for deletion budget and `R` for preservation loss.
6. **Evidence and relevance.** Is the synthetic batch a useful illustration of exact-target gaps given its eligibility rate and small parameter range? What additional weighted instances, controlled `R` scaling, or generic ILP baseline would materially test practical value?
7. **Publication positioning.** If the proof is correct, is the specialized parameterized result sufficiently distinct and substantial for the intended venue? Which precise claim needs narrowing or which closest source is still missing?

A useful response can be short: **claim assessed; verdict; theorem/page or counterexample; required revision**. No reviewer is being asked to endorse originality without checking the relevant result.

Reproduce the lightweight checks on a Mac

From the repository root, with Python 3.8 or newer:

```
cd temporal-profiles
python3 verify_artifacts.py
python3 implementation/demo.py
python3 analysis/summarize_batch.py results/BATCH_RESULTS.zip
```

These commands check integrity, run the bundled examples, and summarize saved results. They need no IBM dataset and do not rerun the original Mac batch. The full reproduction guide documents the optional exhaustive suite and dataset run.

Version and provenance

The comparison uses repository base commit `cc03ddb6454f00378fae2f5570144f741090fe87`. The scientific snapshot remains unchanged; the companion is outside its frozen checksum manifest.

Artifact	SHA-256
Named manuscript PDF	`13bd3266bb3a045d5acf6e45b95667f18f69fa8ee2b44451abd1377630021941`
Reference solver	`a9f8f2c9f276055ab570940d784244192d177c16299f7a295ad75c481de7f4ba`
Native batch export	`cd16912a10fe1f1e8444824f5a0648b57f2c2b05133e50be5bb1f24d522204c2`

Goutam Dyapa ran the reported macOS experiments. ChatGPT assisted with manuscript drafting and revision, code development, and validation tasks, as disclosed in the manuscript. ChatGPT also assisted in preparing this review companion. This disclosure does not imply independent human review.

Novelty comparison

Goutam Dyapa | 12 November 2026

Companion to the 5 November 2026 working manuscript. Reviewer guide | Combined PDF

1. Assessment and closest results

The defensible candidate is a specialized algorithmic theorem. The problem can be connected to a broad existing deletion-propagation framework. Computing a full exact profile, using strict temporal paths, or using a tree dynamic program is insufficient by itself to establish originality. The review should test the particular sufficient state, preservation-loss charging, configuration bound, and resulting time and space guarantees.

This is a focused comparison with four primary research sources, not an exhaustive novelty search. It neither proves that Theorem 2.1 is new nor establishes that an existing theorem implies it. The derived construction below makes the formulation overlap concrete.

Prior work and exact location	Established overlap	What still needs an argument
Makhija and Gatterbauer, GDP [1], Definition 4.1, p.2670; Section 5, pp.2672-2676	Multiple views can have simultaneous deletion and preservation requirements; a general ILP handles their shared source decisions. Self-joins and set/bag semantics are supported.	Whether a theorem for the encoded joint problem yields the manuscript's preservation-loss bound. Binary event costs are an explicit ILP extension here, not a claim about the literal cardinality objective in Definition 4.1.
Molter, Renken and Zschoche [2], Theorems 14-15, pp.10-12	Polynomial forest reachability optimization; strict traversal is included. The weighted delay model has reached-vertex weights and protected contacts. Its DP uses first-arrival information and min-plus budget combination.	A transfer must preserve exact tuple multiplicities, shared event decisions, costs and the relevant parameter. Reached-vertex weights do not directly equal deletion costs.
Enright, Meeks and Molter [3], Theorem 2, p.754	Temporal path counting on forests with `n` vertices takes `O(n * lifetime^2)` time. The manuscript's Proposition B.1 gives its fixed-deletion counting transfer.	Counting after a chosen deletion does not select a minimum-cost shared deletion set or establish the stated optimization bound.
Hu, Sun, Patwa, Panigrahi and Roy [4], Definition 3.1, p.230	Aggregated deletion propagation removes at least a requested number of answers to one query, minimizing the number of deleted source tuples.	This older formulation is narrower than GDP. Its limitations cannot be used to claim that GDP lacks simultaneous exact counts or preservation.

The final reachability paper is **Journal of Computer and System Sciences 144 (2024), article 103549**. Its forest theorems are 14 and 15; the MFCS 2021 predecessors are 12 and 13. Its Theorem 16, p.13, concerns non-strict traversal and timed feedback vertex number. These version and parameter distinctions matter.

The existing manuscript already credits the reachability and counting results in Section 9.1. The substantive addition in this companion is the GDP construction and the resulting narrower positioning.

2. Derived GDP encoding: exact counts are expressible

The following is a construction developed for this comparison, not a temporal example claimed by the GDP authors. It uses ordinary relational queries and therefore does not assume that timestamp inequalities are primitive syntax in their definition.

Preserve event identity and strict timing

Apply forced deletions first and reserve their total cost. Let `E0` be the remaining events. Put one row per original ID into a unary relation `Event(id)`.

Precompute two fixed compatibility relations from the baseline history:

- `H2(i,j)` contains exactly the valid strictly increasing, root-ending two-event ID tuples.
- `H3(i,j,k)` contains exactly the valid strictly increasing, root-ending three-event ID tuples.

Their sizes are at most `M^2` and `M^3`, where `M` is the explicit event count. Precomputation performs comparisons on binary-encoded timestamps; it does not expand the numerical lifetime into a time grid. Protect every row of each compatibility relation by preserving its identity view at full cardinality.

Define the queries:

```
Q2(i,j)   :- H2(i,j),   Event(i), Event(j).
Q3(i,j,k) :- H3(i,j,k), Event(i), Event(j), Event(k).
```

Every output retains all event IDs. Thus set semantics counts each distinct event tuple once, even when endpoints and timestamps are duplicated. Deleting one `Event` row simultaneously affects every tuple containing that event. For each mandatory keep event, preserve its singleton `Event` view. Forced deletions have already been removed from the baseline.

Enforce one exact profile entry

For a candidate pair `(w2,w3)` with `T0 <= w2 <= N2` and `0 <= w3 <= N3`, impose both requirements for each `h` in `{2,3}`:

```
Delete at least N_h - w_h outputs of Q_h.
Preserve at least w_h outputs of Q_h.
```

Since deleting input rows can only remove these outputs, the two inequalities force exactly `w_h` survivors. Omit any requirement with threshold zero, and omit protection for an empty compatibility relation; these impose no restriction and the published definition uses positive requirement vectors. Putting a query in both the deletion and preservation lists is allowed by the GDP framework. Minimize deletions in the `Event` identity view for the unit-cost case. Each feasible solution then corresponds to a legal original-event deletion set with the requested counts, and conversely.

Recover the whole profile table

Solve for every candidate pair and retain each optimum and one attaining event set. There are at most `(R + 1)(N3 + 1) = O((M + 1)^5)` pairs. The compatibility relations and each instance have polynomial size. This derives the full unit-cost table through polynomially many GDP calls. It does **not** bound the time or memory needed to solve those calls in terms of `R`.

Consequently, simultaneous exact counts, preservation, identity-sensitive multiplicity, forced choices and full-table output are not sufficient separations from GDP expressibility. The restricted temporal structure may still enable a new algorithmic guarantee.

3. Weighted ILP extension and a useful diagnostic

Direct witness formulation

Let `y_e` be a binary variable indicating that a baseline event survives. Fix `y_e = 1` for mandatory keep events. For every tuple `q` in `H2` or `H3`, let binary `z_q` indicate that the tuple survives. Impose:

```
z_q <= y_e                          for every event e in q
z_q >= sum(y_e for e in q) - |q| + 1
sum(z_q for q in H2) = w2
sum(z_q for q in H3) = w3
```

Minimize:

```
cost(forced deletions) + sum(c_e * (1 - y_e) for e in E0)
```

These inequalities make `z_q` the conjunction of the retained event decisions. They prevent double counting when several deleted events destroy the same tuple. All coefficients `c_e` retain their original binary encoding; no event is copied once per unit of monetary cost. Zero costs are permitted.

This is a transparent weighted extension of the witness-indicator approach. The literal GDP Definition 4.1 uses cardinalities in its objective. The weighted construction should therefore be presented explicitly, without claiming that its full binary-cost specification is already a stated GDP theorem. No ILP implementation or performance benchmark is supplied by this companion.

Why generic LP integrality cannot be assumed

Consider two distinct unit-cost middle events `b1,b2`, both at time 0 from `v` to `u`, and one unit-cost final event `c` at time 1 from `u` to root `z`. This is an eligible depth-two history. Its baseline is `(N2,N3) = (2,0)`. Set `T0 = 1`, so `R = 1`, and ask for exact `(W2,W3) = (1,0)`.

An integral solution must delete exactly one middle event; its optimum cost is 1. Deleting the final event leaves zero two-walks and misses the target. In the linear relaxation of the displayed formulation, the assignment

```
y_b1 = y_b2 = 1;  y_c = 1/2
z_(b1,c) = z_(b2,c) = 1/2
```

satisfies every displayed constraint with objective `1/2`. This is an algebraic diagnostic for this formulation. It rules out automatically treating its LP relaxation as integral on all eligible histories. It does not rule out stronger formulations, cutting planes, other GDP algorithms, or another polynomial method.

The parameter transfer must also be proved

Take `n` optional unit-cost first events before one mandatory-retained middle/final pair, preserve that pair, and require zero triples. Then `R = 0`, while all `n` first events must be deleted. Thus a bound parameterized by deletion count need not become a bound parameterized by `R`.

GDP Section 6, Propositions 6.1-6.4, recovers particular tractable deletion-propagation variants. This comparison has not derived the manuscript's `2^{O(sqrt(R))} poly(L)` time and polynomial-space guarantee from those results. A formulation encoding and an algorithmic implication are separate claims requiring separate evidence.

4. Positioning and review priorities

Suggested contribution wording

The manuscript studies a structured temporal deletion problem and claims an exact algorithm parameterized by the allowed loss of two-event walks. On explicit depth-three in-arborescences, its final-event grouping and chronological state yield every attainable minimum-cost two-/three-walk profile in `2^{O(sqrt(R))} poly(L)` time and polynomial space. The formulation relates to generalized deletion propagation; the candidate contribution is the specialized structural algorithm and complexity guarantee, together with the stated deletion-budget hardness result.

This wording should remain conditional on proof assessment and a sufficiently broad literature comparison. It avoids claiming to introduce exact deletion constraints, temporal counting, tree dynamic programming, or min-plus convolution.

What would settle the remaining questions

Priority	Concrete evidence needed
Proof review	An expert check of Lemmas 3.1-3.3 and 4.1, all Section 5 complexity accounting, and Appendix A's reduction in both directions.
Theorem comparison	A theorem-level search for prior ordered-incidence, query-optimization, or temporal algorithms implying the same bound; any proposed transfer must preserve costs, shared decisions and its parameter.
Practical comparison	A tested generic ILP baseline and controlled instances varying `R`, event count, branches, timestamp ties, and cost bit length, if practical efficiency is to be claimed. The current batch does not establish a speed advantage.
Manuscript revision	Incorporate this GDP relation, cite the primary paper, qualify the weighted extension, and focus the novelty statement on the specialized theorem. Keep the frozen snapshot distinct from that future revision.

The reachability paper's first-arrival state alone does not determine tuple multiplicity: manuscript Appendix B.2 gives equal-reachability histories with different `w3`. That is a limited obstruction to a direct summary, not a proof that no reduction exists. Likewise, polynomial counting of a fixed retained history does not itself optimize deletion choices. These observations identify work that a transfer argument must do; they do not certify novelty.

References and source scope

[1] Neha Makhija and Wolfgang Gatterbauer. *Is Integer Linear Programming All You Need for Deletion Propagation? A Unified and Practical Approach for Generalized Deletion Propagation.* PVLDB 18(8), 2667-2680 (2025). DOI 10.14778/3742728.3742756. Primary PDF. Author full version, arXiv:2411.17603. Published Definition 4.1 and Sections 5-6 were the main comparison targets.

[2] Hendrik Molter, Malte Renken and Philipp Zschoche. *Temporal reachability minimization: Delaying vs. deleting.* Journal of Computer and System Sciences 144, article 103549 (2024). DOI 10.1016/j.jcss.2024.103549. The supplied final journal full text was inspected. The earlier MFCS 2021 open version has different theorem numbers.

[3] Jessica Enright, Kitty Meeks and Hendrik Molter. *Counting Temporal Paths.* Algorithmica 87, 736-782 (2025). DOI 10.1007/s00453-025-01301-3. Author-hosted final article. Theorem 2 is on printed p.754.

[4] Xiao Hu, Shouzhuo Sun, Shweta Patwa, Debmalya Panigrahi and Sudeepa Roy. *Aggregated Deletion Propagation for Counting Conjunctive Query Answers.* PVLDB 14(2), 228-240. DOI 10.14778/3425879.3425892. Primary PDF. This comparison uses Definition 3.1; the volume/pages and DOI identify the article despite differing online/issue year conventions.

The broader sources already discussed in manuscript Section 9, including set-cover and other optimization frameworks, remain relevant. This companion does not replace that bibliography or claim to have eliminated every potential subsumption route. No correspondence or approval from the cited authors is assumed.